Introduction

AI-Powered Smart Waste Management System

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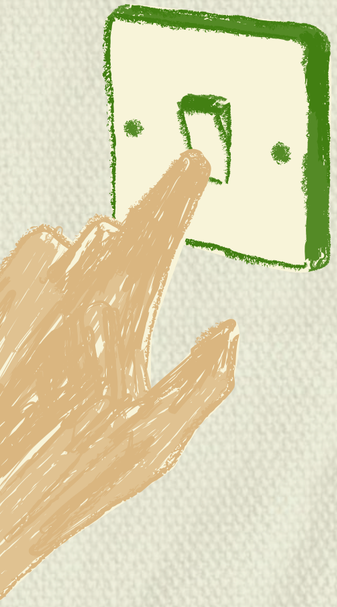
PROBLEM

- Inefficient waste collection system
- High fuel & operational cost
- No prediction mechanism
- Low citizen engagement
- Lack of real-time monitoring

Existing Solution

- Fixed route collection
- Manual complaint handling
- No AI prediction
- No citizen reward system



OUR DIFFERENT SOLUTION:

- ✓ Software-only system (No IoT hardware required)
- ✓ AI prediction model
- ✓ Citizen behavior scoring
- ✓ Gamification approach
- ✓ Scalable cloud deployment

TECHNOLOGY USED

BACKEND

- Node.js, Express.js
- Database

PostgreSQL

- AI / ML

LANGUAGES

- JavaScript

Python

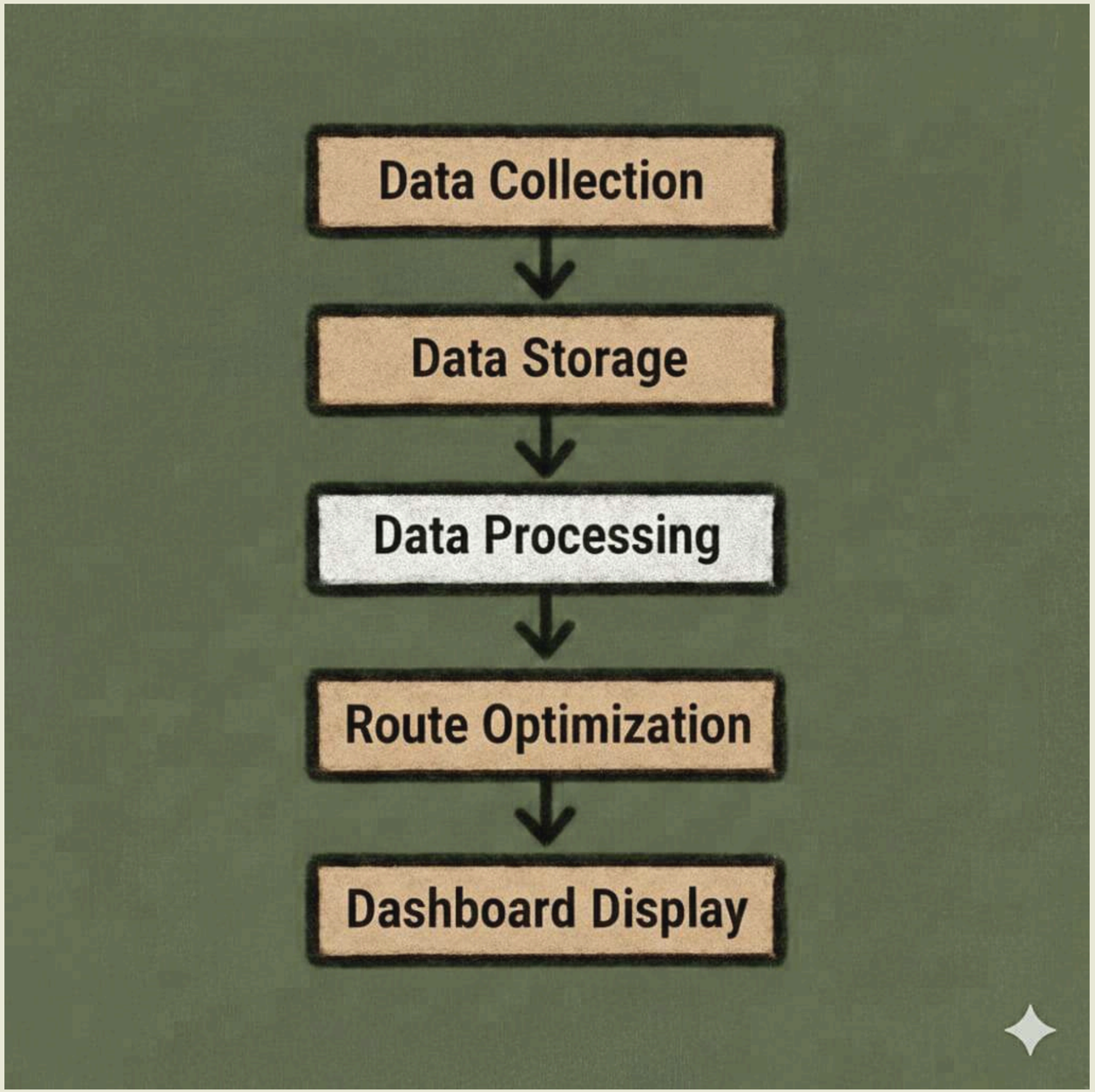
- CloudCloud

AWS / Google Cloud

FRONT END

- React.js

FLOW CHART


Data Collection
Data Storage
Data Processing
Route Optimization
Dashboard Display

METHODOLOGY

- Collect historical waste data
- Preprocess data
- Predict waste generation
- Generate routes
- Update dashboard
- Monitor & improve system

Modules

Waste Prediction Module

Route Optimization Module

Citizen Green Score Module

Admin Dashboard Module

CONCLISION

“Our AI-driven software solution builds cleaner, smarter and sustainable cities.”